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Review article

Internet of Things (IoT) security dataset evolution: Challenges and future directions

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ABSTRACT

The evolution of mobile technologies has introduced smarter and more connected objects into our day-to-day lives. This trend, known as the Internet of Things (IoT), has applications in smart homes, smart cities, industrial automation, health monitoring systems, and has become an essential component of the communication and networking industry. However, different communication and protocol standards, weak security defaults and the difficulty of distributing updates have exacerbated cybersecurity threats to critical applications that employ IoT. To mitigate the threats and counter these attacks, a promising approach is to develop a robust intrusion detection framework specifically aimed at securing IoT. This paper presents our efforts to catalogue and compare attacks, datasets and machine learning algorithms and architectures for intrusion detection systems for IoT devices. We classify attacks aimed at IoT devices at different layers and protocols. This work also highlights potential features that can be used by machine learning-based intrusion detection systems to detect different types of attacks. We provide a comparative study of IoT datasets used for model training and identify key properties which helps in assessing their suitability in particular scenarios. Finally, we discuss our observations and propose the research directions for building a robust IoT intrusion detection system.

1. Introduction

Advancements in mobile technologies have paved the way for the emergence of the Internet of Things (IoT), revolutionizing, for example, healthcare, homes, and cities with smart infrastructure [1]. These smart devices can be seen as interconnected parts and are built with network interface cards and lightweight processors and managed through various interface services (e.g., web pages, graphical user interface, and remote login) [2]. With an ever-increasing demand for IoT devices, new and unique threats are affecting these devices. In this sense, an important step is to develop a forensic IoT methodology to capture and investigate the adversarial behavior and identify the role of devices in performing cyberattacks [3]. With IoT networks generating an enormous amount of data, analytical techniques are generally incapable of handling this data in real-time.

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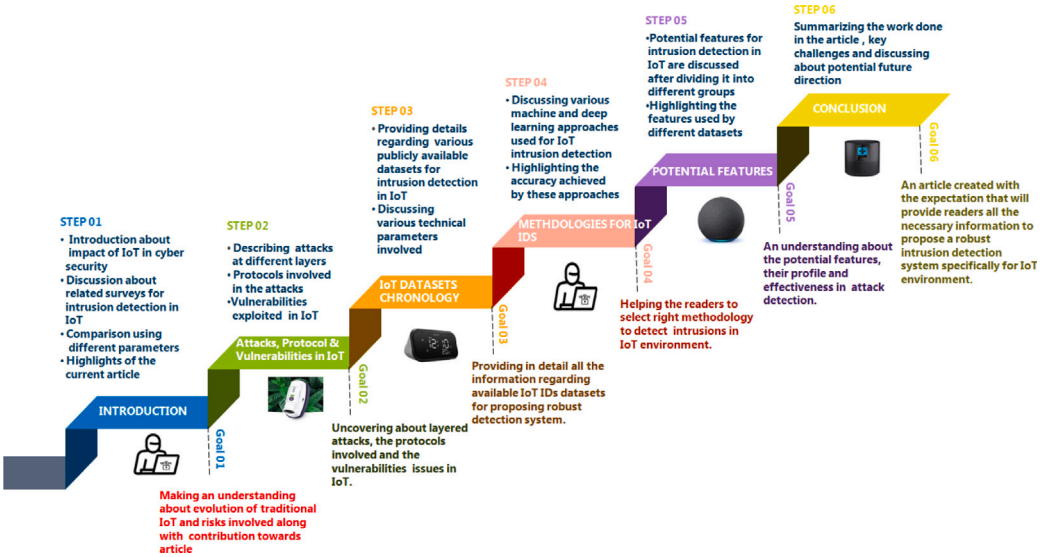

Fig. 1. A stair-step model of this article showing different steps and goals achieved after completing each step.

Table 1
A comparison of the state-of-the-art surveys.

Survey	IoT layered attacks	IoT attacks in-depth	IoT protocols	IoT vulnerabilities	IoT features	IoT detection techniques		Accuracy	IoT datasets
						DeepLearning Methods	Traditional Methods		
H.Hindy et al. [2018] [10]	✓	✓	•	•	•	◎	✓	•	•
Ring et al. [2019] [11]	◎	◎	•	•	•	•	•	•	•
Khraisat et al. [2019] [12]	◎	•	•	•	•	•	◎	•	◎
MA Ferrag et al. [2020] [13]	•	•	•	•	•	✓	•	•	◎
J Asharf et al. [2020] [14]	✓	•	✓	•	•	✓	✓	•	◎
Dilara et al. [2020] [15]	•	•	•	•	•	✓	•	•	◎
khraisat et al. [2021] [16]	◎	◎	•	•	•	✓	✓	•	◎
Hamid et al. [2021] [17]	•	•	•	•	•	•	•	•	◎
Aversano et al. [2021] [18]	•	✓	•	◎	◎	✓	✓	•	•
Lohiya et al. [2021] [19]	•	•	◎	•	•	✓	✓	•	•
This survey	✓	✓	✓	✓	✓	✓	✓	✓	✓

(✓) Fully discussed, (◎) Partially discussed, (•) Not discussed.

Traditional IoT systems have static defense mechanisms in the form of firewalls, anti-viruses, or software patches, which has become a constraint for the device manufacturers due to the heterogeneity of such devices [4]. Furthermore, the multi-functionality factor has also posed the challenge of developing and deploying a security mechanism capable of detecting attacks over a wide range of devices. This has resulted in the in-feasibility of robust security measures [5]. The widespread deployment of IoT applications means that their security has become a vital element for secure communications [6]. Also, with the advancement in Artificial Intelligence (AI), many deep and machine learning approaches are being proposed to improve intrusion detection systems (IDS). Several survey papers have highlighted the different traditional benchmark IDS systems proposed [7,8].

However, the existing studies do not offer any insight into the latest developments. Current survey papers also evaluate the threats at different layers in IoT systems by comparing tools available for simulations and their roles in the existing scope of IoT architecture layers. However, with new protocols being proposed and used at different layers, the role of attack and data recording tools were not discussed in such studies [9].

We stress that the access to complete information is crucial to building a robust IDS for IoT environments. New IoT devices are attached to the internet every second, and existing devices and methodologies are becoming obsolete. Therefore, security researchers should have access to an investigation that touches every aspect of the latest developments related to IoT security, which could help detect zero-day exploits and defend against new threats. Fig. 1 depicts a stair-step visualization of our contributions where each "step" highlights the topics discussed in the paper, and the "goals" define what the reader will learn before going to the next step.

1.1. Review of previous surveys

Recent past research shows that numerous IDS initiatives were developed and evaluated using public datasets or datasets provided on request. Several studies were reviewed and have been published, highlighting the security of IoT across different domains.

The survey work presented in [10] mainly focuses on various types of intrusions in IDS while providing every detail. The authors also provided a taxonomy of the attacks, including the layers which are most impacted by these attacks. The attack tools are explained in great detail. But the article does not give direction about the algorithms that can be used to detect the attacks for IDS nor discuss available benchmark datasets in detail to check the system's efficacy. An exhaustive literature survey on the NIDS datasets covering a variety of features, formats, and relationships between datasets is performed [11]. The paper also discusses attack scenarios, traffic generators, recording environments, and data repositories. It provides some recommendations for the researchers to make decisions based on their scenarios, i.e., the study discussed 34 datasets and identified 15 different characteristics. However, the number of samples in each attack for every dataset has not been discussed in detail. Furthermore, the protocols on which the attacks are performed and the models used to detect those attacks are also not presented.

A survey presenting contemporary IDS in the form of taxonomy is reviewed [12], citing notable works while discussing the datasets commonly used for evaluation purposes. In this paper, different evasion techniques for detecting attacks are also surveyed with future research challenges. The same authors [16] presented a comprehensive survey paper discussing recent notable works related to various intrusion detection datasets. They also presented a taxonomy of IDS, mostly highlighting types of attacks based on intrusion methods. Different performance evaluation metrics and the importance of feature selection are also mentioned while discussing the challenges of evasion techniques. Although their study discusses the datasets related to attacks in the network and partially covers the IoT environment, there is a lack of details about the features used to detect those attacks and the accuracy achieved by the models for detecting those attacks.

Various deep learning approaches for intrusion detection can be broadly categorized into two different models: deep discriminative and generative/unsupervised. The authors also categorized 35 different cybersecurity datasets into seven categories. Two real traffic datasets, CSE-CICIDS2018 and Bot-IoT, have been used to evaluate the binary and multi-class performance of the proposed models. However, there is a lack of discussion about other available IoT datasets [13].

The authors in [14] discussed different IoT architecture layer attacks with threat dimensions and attack surfaces, including the technology, frequency bands, and respective data rates, and highlighting various available network and IoT-related datasets. However, they do not discuss the features and approaches which help precisely detect these attacks.

A review analyzing the recent methods for intrusion detection is presented in the form of deep learning methodologies. A detailed analysis of the intrusion detection benchmark datasets has also been conducted, which shows the usage of deep learning models for training. The authors partially discussed the IoT-related dataset but did not provide any information regarding the attacks performed [15].

A recent review article based on eleven different contributions investigated several relevant IoT-based botnet attacks. The literature paper partially discussed the available IoT datasets. However, Hamid et al. [17] followed characterization techniques to detect and mitigate such attacks. This study has not discussed the IoT device name, the type in which the attacks were performed, or the algorithms used to detect the attacks. Finally, the works presented in [18] and [19] bring insights on the application of ML and DL in IoT security, but do not focus on IoT vulnerabilities and layered attacks.

The review papers discussed so far consider only specific aspects of the datasets, i.e., they do not cover detailed requirements about IoT-based environments. A solid foundation is required before proposing IDS for IoT. This paper offers a comparative study of the available IoT datasets and techniques followed to detect the attacks and highlights the need to have an up-to-date and robust IDS for IoT environment. Table 1 compares the state-of-the-art surveys concerning IoT datasets, features, models, and accuracy achieved.

1.2. Contributions

This paper helps researchers identify the current IoT datasets for intrusion detection and their characteristics. The following are the main contributions of this paper:

- We analyze some of the current state-of-the-art surveys providing information regarding which survey covers IoT dataset information fully or partially;
- We investigate different security attacks in the IoT environment by highlighting the taxonomy of each attack. Security aspects (e.g., confidentiality, integrity, availability, the impact of the threats, and privacy) are considered. Details of different attacks on protocols are discussed in-depth alongside the exploited vulnerabilities from which IoT suffers;
- Several well-known publicly available IoT datasets are analyzed and compared by identifying fundamental properties, i.e., domain, availability, kind of traffic, data format, labeled/non-labeled, duration, network type, country, number of devices, number of samples for observation and evaluation purposes;
- An extensive discussion on future directions is conducted throughout the paper and summarized in the final section. The motivation for this initiative relies on fostering the development of new initiatives to benefit the scientific and industrial communities. In this sense, gaps in the current state-of-the-art are presented covering different IoT security aspects.

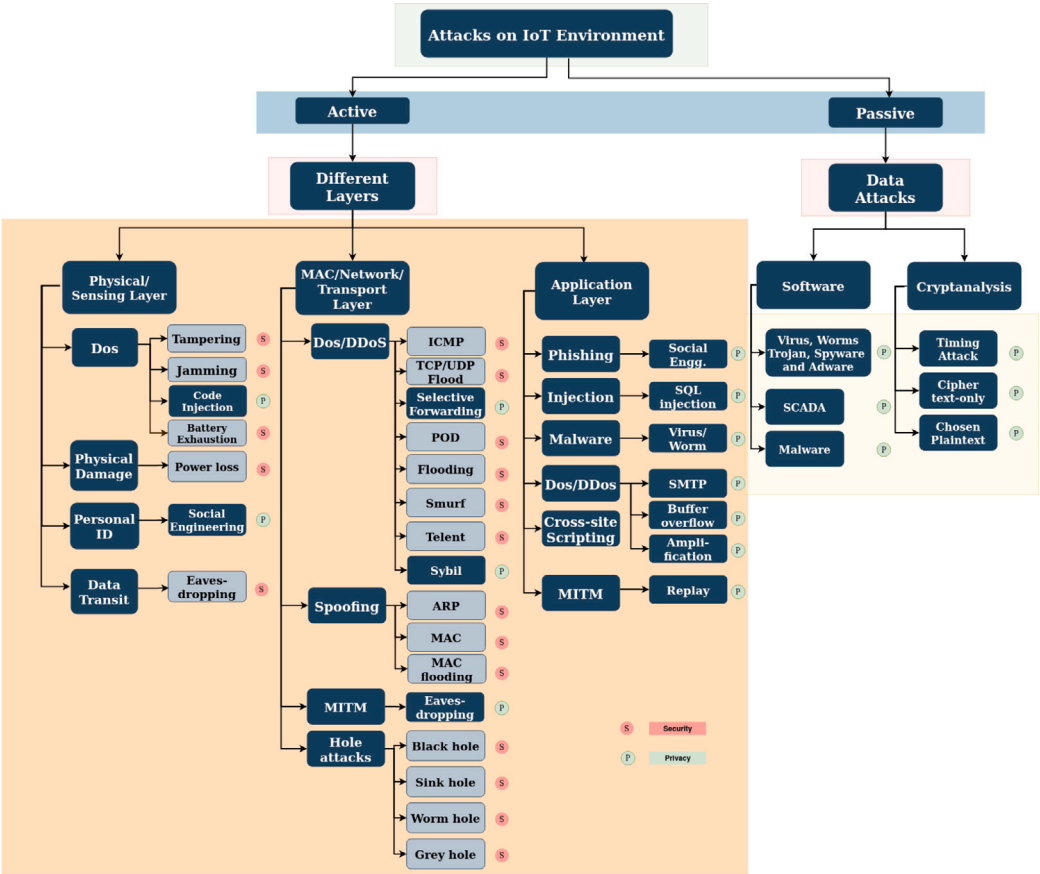

Fig. 2. Taxonomy of attacks on IoT performed at different layers.

Overall, this paper contributes to researchers interested in proposing or developing a robust IoT-based intrusion detection system and significantly impacting this upcoming research arena. The rest of the paper is organized as follows: In Section 2, well-known attacks at different layers are discussed with in-depth attack taxonomy included; attacks on protocols and vulnerabilities exploited are also highlighted. In Section 3, various available IoT related datasets are presented with properties in detail. We discuss the techniques used by the datasets to detect the attacks in Section 4. In Section 5, the list of potential features used by the datasets are provided in details. Finally, a conclusion of the work with future research directions is presented in Section 6.

2. Security attacks in IoT

In the past decade, IoT devices have been facing different kinds of attacks, making users conscious of using these devices more carefully. The interest of the attackers is either information gathering, data theft or denial of service to the legitimate users. The main objective of this section is to determine actual attacks and examine respective targets by dividing the attacks into two broad categories, active and passive attacks. We further highlight different layers of network models that are affected by these attacks. We also discuss different protocol attacks and exploited vulnerabilities related to the IoT environment. Fig. 2 illustrates our proposed taxonomy of attacks on IoT.

2.1. Active attacks

In this type of attack, the intruder breaks into the system and alters the corresponding information. It attempts to control the resources by introducing malicious code, disrupting the normal operations by transmitting bulk information, saturation and disseminating the information [20]. For example, the Denial of Service (DoS) can start dropping data packets and consuming resources [21]. The active attacks are defined layer-wise in the following section.

2.1.1. Physical layer attacks

Another name for this layer is the sensing layer, where various channels are used for communication among the devices. The physical attacks range from unauthorized access to the device to the removal or blocking of the connections of legitimate users. The attacker causes physical damage to the device by not letting the system shut down, ultimately leading to overheating [22].

2.1.2. MAC/network/transport layer attacks

As MAC addresses are coded into network interface cards, cards, the attacker launches a spoofing attack by changing legitimate users' MAC addresses. Other threats like Man-in-the-Middle (MITM) attacks can prevent the working of specific nodes on a network [23].

At the network layer, a significant amount of packets are injected, causing the network to be congested. This depletes the resources throughout the network. Also, most of the attacks are performed on the IP protocol due to efficient delivery of packets from one point to another, thus providing easy access to the service at this layer.

2.1.3. Application layer attacks

This layer supports several protocols with different functionality; for instance, SMTP is used for mail transfer, while HTTP enables web services. IoT device usage in different domains like smart housing and office application layer protocols, provides an easy and accessible ground for cyber attackers.

2.2. Passive attacks

This type of attacks usually gather information without consent from the user and further exploit the information by decrypting it. Passive attacks are further categorized into the following attacks:

2.2.1. Software

Usually, malicious software is used to infect the system for stealing the data system to steal data or even to launch DoS attacks. This software could in such forms as viruses, worms, and spyware. The SCADA systems are the most vulnerable to attacks due to their connectivity to the IoT network. Due to the exponential growth in the IoT technology sector, conventional SCADA systems are integrated with cooperating business networks and IT systems.

2.2.2. Cryptanalysis

Several lightweight cryptographic algorithms have been developed to meet the increasing demand for IoT devices that use extremely low resources without human intervention. A vast amount of raw information is recorded daily from devices enabled with IoT As most commonly used algorithms use significantly less computing resources, the security of IoT devices using lightweight cryptographic algorithms is more vulnerable to cryptanalysis techniques such as differential cryptanalysis or slide attacks. Providing security against these attacks to IoT devices, which are connected at different layers, has become crucial. Some techniques or strategies must be used to counter the attacks when working with the IoT model.

2.2.3. Volume based and bandwidth/resource depletion attack

Here, the attacker utilizes all the bandwidth by sending bots over the network, preventing legitimate users from accessing allocated services. Also, the attacker starts depleting the resources by allocating or depleting them. Different layers of network architecture are targeted. This attack is severe, causing systems to crash due to factors such as memory and CPU usage overflow.

2.2.4. Web based attack

The attacker forwards numerous requests to a particular website on the web, making it non-functioning for many hours. Also, eavesdropping is performed by intruders to capture personal data of the users.

2.3. Attacks on different types of protocols

The protocols are designed to provide effective communication from source to destination. Moreover, these protocols usually work in a simple yet collaborative environment not concerned with integrity and privacy mechanisms. Due to these reasons, the protocols working at different layers are more vulnerable to attacks [25]. In Fig. 3, A fishbone diagram, popularly known to represent security and safety, has been used for categorizing individual attacks on protocols. These diagrams provide a thoughtful analysis of the possible contributing factors (or sub-causes) of a particular problem [24]. In fact, different protocols and threats are considered.

2.3.1. Attacks on internet protocol (IP)

The IP protocol works at the network layer, whose functionality is to route the packets from the source to the destination following the router rules. Here every next hop is determined by the router, which ensures that packets reach their destinations. Several attacks can occur against the IP protocol.

2.3.2. Attacks on TCP

The functionality of this protocol is to provide guaranteed delivery of the packets over a network. The TCP protocol usually follows a mechanism of transmitting a packet [26]. But the attacker intrudes into the system by initiating many half connections [27] performing TCP-SYN flood attack, bypassing the firewalls that do not block FIN packets [28], session hijacking [29].

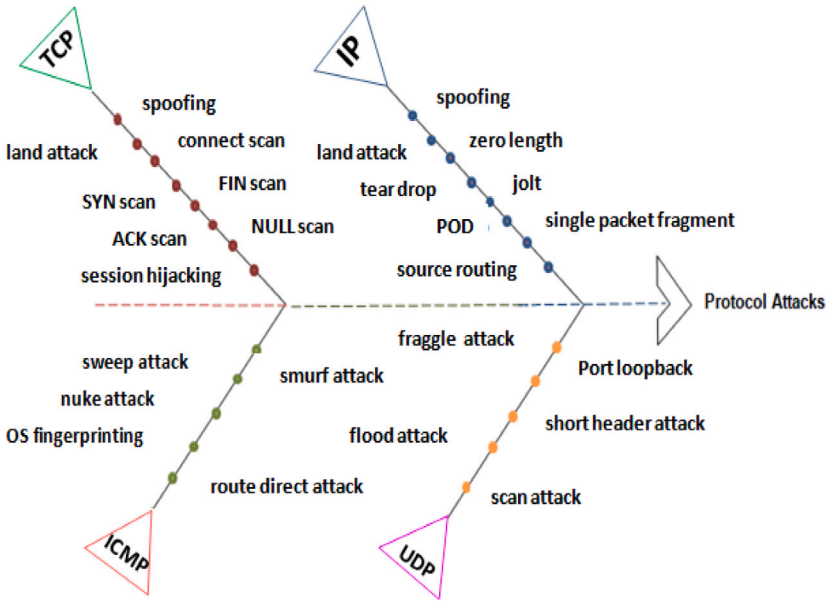

Fig. 3. A fishbone model for categorizing attacks concerning protocols [24].

2.3.3. Attacks on UDP

An interface for the IP is the User Datagram Protocol (UDP), whose function is to transmit a datagram from one application to another with low overheads. However, due to its nature, UDP is always vulnerable to different attacks [30], other attacks include port loopback, fraggle attack [31].

2.3.4. Attacks on ICMP

This protocol sends a one-way message used by the routers in the network, mainly information regarding errors related to data processing by the destination hosts. During this process, many attacks (e.g., address sweep, smurf attack, and OS fingerprinting attack) are based on sending malformed ICMP packets to the destination [32].

2.4. Type of exploited vulnerability

A vulnerability is a weak area where an attacker finds an easy way to access the system. The increasing variety of IoT devices led to more vulnerabilities and cyber attacks. Common vulnerabilities related to IoT devices are weak passwords [33], open ports [34], insecure application interfaces [35], exposed network interface [36], exposed software [34], device memory, insecure cloud interface [36], password cracking, underlying infrastructure security, and virtualization threats [37,38]. Table 3 highlights the types of vulnerability and attack types.

Moreover, IoT-based networks are different from traditional networks in several ways, as shown in Table 2. In fact, there is an open research problem regarding how the internal communication pattern can differ and how Machine Learning (ML) can be used to underpin such patterns. For example, IoT profiling [6,39] is a current research problem that is focused on the communication and exchange of IoT devices, involving how Machine Learning (e.g., Deep Neural Networks and Random Forest) [40,41] and anomaly detection (e.g., Isolation Forest) [42] methods can be used more accurately from interpretability efforts [43]. Finally, the granular analysis of the intersection between features of IoT-based networks, IoT profiling and behavioral aspects in IoT-based attacks, and how ML techniques underpin single and multiple threats, is included in the future directions of this research.

Different Machine learning methods can recreate a critical role in IoT security by providing enhanced threat detection, anomaly detection, and predictive analytics capabilities. The following represent some of the important ML techniques for detecting different threats in IoT environment:

- Anomaly detection: Anomaly detection is a technique that involves identifying abnormal or unexpected behavior in an IoT-based network. IoT can include monitoring sensors and devices for unusual behavior that could indicate a security breach or threat. Machine learning algorithms can be trained to recognize these anomalies and alert security personnel to investigate further. Using either nearest neighbor, cluster-based, classification-based, or statistical-based anomaly algorithms such as k-NN, Local Outlier Factor (LOF), and Connectivity-based Outlier Factor (COF) to be trained based on different normal behavior of the devices such as their time interval to connect to the sensor, connection time for sending ARP request or other network protocol behavior of the IoT devices shows how vital a machine learning can be in IoT environment, especially in preventing an attack. Furthermore, installing a lightweight anomaly algorithm on your IoT-based network can save the organization from timely attacks;

Table 2
Comparison of traditional and IoT-based networks [44–46].

	Traditional networks	IoT-based networks
Data generation	Interaction of multiple systems	Automated procedures trigged by sensors and actuators
Topology and Services	Centralized	Decentralized
Security	Robust Security Mechanism	Lightweight Security Mechanisms
Adaptability	Stable and Rigid Topologies	Highly flexible and adaptable
Computation	Processing Power Available at end nodes	Processing power provided by remote systems
Communication	Supports complex protocols	Prioritizes Lightweight Protocols

Table 3
Common type of attacks along with vulnerability.

Type of attack	Vulnerabilities												
	Weak passwords	Open port	Data confidentiality	Insecure application interfaces	Missing privileges	Exposed network interfaces	Exposed software	Node tempering	Insecure cloud interface	Device memory	Infrastructure security	Virtualization threat	Password cracking
Aidra [47]				✓									
Backdoor [48]		✓										✓	
Bashlite [49]					✓								
Botnet [49]								✓					
Brute Force [50]	✓		✓	✓			✓						✓
DDoS [51]					✓			✓					
Doflo [52]					✓								
DoS [47]								✓	✓	✓			
Data Theft [52]				✓			✓				✓		✓
Hajime [47]								✓					
Mirai [49]	✓							✓	✓				
Info. Gather [52]			✓	✓					✓			✓	✓
MITM [53]	✓	✓				✓							
Reconnaissance [51]		✓											✓
Ransomeware [48]	✓		✓	✓	✓	✓			✓				
Slowlte [50]						✓							
Smurf [54]													
Spoofing [23]			✓			✓			✓				
Scanning [52]		✓											
Sniffing [55]						✓			✓				✓
Tsunami [47]					✓								
Trojan [56]				✓		✓							
Torii [47]			✓										
Virus [57]	✓						✓				✓		
Web [48]			✓	✓			✓			✓	✓	✓	✓
Worm [58]	✓						✓				✓		

- Predictive analytics: Predictive analytics uses historical data to identify patterns and trends that can help predict future behavior. In IoT security, predictive analytics can be used to anticipate potential threats and take preventive measures before they occur;
- Deep learning: In IoT security, deep learning can be used to analyze large amounts of data generated by sensors and devices in real-time, detecting security threats and anomalies more quickly and accurately.
- Reinforcement learning: In IoT security, reinforcement learning can help devices learn to identify and respond to security threats in real-time.

Machine learning is vital in IoT security by providing enhanced threat detection and prevention capabilities. Using the mentioned methods, organizations can analyze vast amounts of data generated by IoT devices and identify potential risks more quickly and accurately.

3. IoT attack datasets

In the past few years, many datasets in the IoT security domain have been assembled, each with pros and cons. Due to the exponential adoption of IoT devices, researchers are increasingly working towards IoT-related datasets because of the upsurge in unknown vulnerabilities and threats. IoT devices' effectiveness, security, and applicability against normal or abnormal behavior are tested by capturing the data in a simulated or realistic environment. Therefore, dataset quality plays a vital role in providing an effective model for detecting intrusion in a real environment. This section discusses publicly available datasets proposed for IDS in IoT environments. Fig. 4 presents the year-wise distribution of IoT datasets proposed, which shows a sharp increase in the year 2020. In addition, Table 4 depicts different datasets with technical details and properties which reflect general information about them.

3.1. Search criteria

In order to find IoT datasets available to support security solutions, three main steps were considered as illustrated in Fig. 5. With the intention of identifying recent datasets published in the past 5 years, we did a search through Google Scholar using the

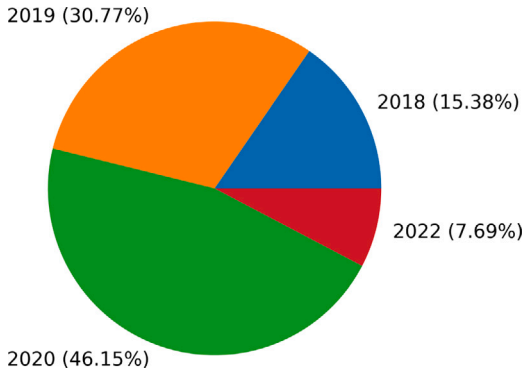

Fig. 4. Number of IoT datasets proposed year-wise.

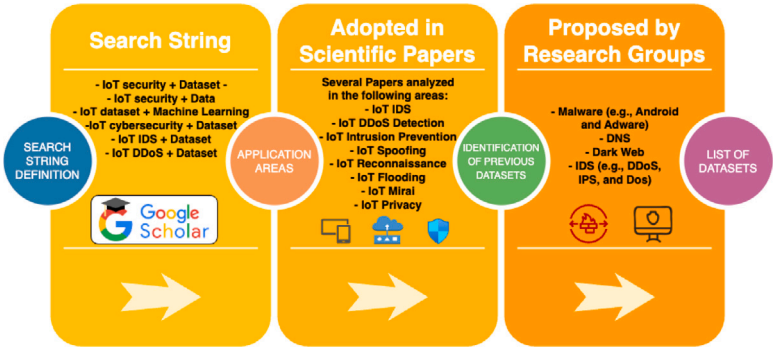

Fig. 5. Search criteria adopted in this research.

following search strings: "IoT security + Dataset", "IoT security + Data", "IoT dataset + Machine Learning", "IoT cybersecurity + Dataset", "IoT IDS + Dataset", and "IoT DDoS + Dataset".

In order to expand our search for datasets available, we also focussed on technical papers that were leveraged by the use of IoT datasets. In this sense, we searched for technical contributions in the following areas: IoT IDS, IoT DDoS Detection, IoT Intrusion Prevention, IoT Spoofing, IoT Reconnaissance, IoT Flooding, IoT Mirai, IoT Privacy.

Finally, we also searched for new datasets of research groups that have published widely used cybersecurity datasets in the past. Our main focus was on searching datasets in the following areas: Malware (e.g., Android and Adware), DNS, Dark Web, IDS (e.g., DDoS, IPS, and DoS).

3.2. Features

Domain: Before capturing data, the target domain needs to be selected. As IoT devices are most common in the home environment and provide an easy testbed for cyberattacks, most recorded data focuses on this domain.

Availability: The viability and quality of the proposed datasets for intrusion detection can only be checked if it is made publicly available for research. The property is marked if it is publicly available or available upon request.

Kind of Traffic: This is an important property, as it helps to evaluate the system on an array of IoT devices and attacks. "Simulated" means data captured within a testbed, and "real" means network traffic captured in the real-world scenario with real devices.

Data Format: Capturing data plays an important role as it further helps in analyzing network traffic. The data can be recorded in PCAP format, which includes payload information and feature format with processed data. Another type is the log file, which collects system-generated records like the operating system or server events.

Evaluation: The label field defines whether the instance collected is normal or malicious.

Total Duration: The time span over which the data is captured is defined.

Network Type: This property is used for evaluation to check adaptability to a specific environment.

Country: This field defines the name of the countries where the data is captured.

Number of Devices: To evaluate the viability of the intrusion detection system, different IoT devices are used in the testbed. This field shows the number of IoT devices included in each dataset.

Table 4
Overview of IoT based Dataset properties.

Dataset	Domain	Availability	Kind of traffic	Data format			Evaluation	Totalduration	Network type	Country	#Devices	#Samples		Reproducibility
				pcap	feature	log	Labeled					Benign	Malicious	
N-BaIoT (2018) [59]	Home	Yes	Real	n.s.	Yes	No	Yes	n.s.	Small	Negev, Israel	9	502595	6506674	PR
BoT-IoT (2019) [52]	Home	Yes	Simulated	Yes	Yes	No	Yes	n.s.	Small	Canberra, Australia	5	9543	73360900	R
Kitsune (2019) [51]	Home	Yes	Simulated	Yes	Yes	No	Yes	n.s.	Small	San diego, USA	9	700000	26472754	R
IoTNIDS (2019) [60]	Home	Yes	Real	Yes	No	No	Yes	1 week	Small	n.s.	2	1756276	1219254	R
MedBIoT (2020) [49]	Home	Yes	Real/simulated	Yes	Yes	No	Yes	6 days	Medium	Estonia, Europe	7	12540478	5305089	R
IoT-23 (2020) [61]	Home	Yes	Real	Yes	No	Yes	Yes	1 year	Small	Czech Republic, Prague	3	30858735	710098146	R
IoTIDS (2020) [23]	Home	Yes	Real	No	Yes	No	Yes	1 week	Small	Oshawa, Canada	2	40073	585710	PR
MQTT (2020) [50]	Home/office	Yes	Simulated	Yes	Yes	No	Yes	1 week	Small	Genoa, Italy	8	11915716	165463	R
MQTT-IoT-IDS(2020) [62]	Home	Yes	Simulated	Yes	Yes	No	Yes	n.s.	Small	Dundee, Scotland	1	1056230	352355	R
TON_IoT (2020) [48]	Home/cities /industry	Yes	Simulated	Yes	Yes	Yes	Yes	n.s.	Small	Canberra, Australia	10	245000	124619	R
IoTHIDS (2018) [47]	Home	On request	Simulated	No	No	No	No	n.s.	Small	n.s	n.s	n.s	n.s	HR
IoT-SH (2019) [53]	Home	On request	Simulated	No	No	No	No	5 weeks	Small	Cardiff, Wales	8	170785	50000	HR
Edge-IIoTSet (2022) [63]	Industry	Yes	Real	Yes	Yes	No	Yes	n.a.	Medium	Guelma, Algeria	10	11223940	9728708	R

Number of Samples (Benign/Malicious): This field describes the number of samples captured for each type, which helps choose algorithms according to the available data.

Reproducibility: describes how reproducible the research is based on several aspects. In terms of reproducibility, datasets can be categorized as Reproducible (R), Partially Reproducible (PR), and Hardly Reproducible (HR). This classification is based on how detailed the data collection is in each contribution as well as how clear it is for other researchers to generate a similar dataset following the described approach and is evaluated as follows:

$$R_s = A + Tr + Pc + Nt \tag{1}$$

Where R_s stands for the reproducibility score and from in $0 \leq R_s \leq 1$, A represents the availability of the dataset (0.25), Tr is the type of traffic used (0.25 for simulated, 0.12 for real), and Pc and Nt represent, respectively, the availability of .pcap files (0.25) and the network type (0.25 for small, 0.12 for medium). In this context, all four components are assigned the same weight since they are complementary to each other and fundamental to validating proposed solutions in the IoT security ecosystem. For example, in order to be used, a dataset must be available to download. Also, if the traffic is simulated, it is simpler to be reproduced in any system. Furthermore, the availability of pcap files enables researchers to access the real traffic and engineer their own features. The network type has direct implications for how a dataset production can be reproduced. Finally, these weights can also be configured in a different way to evaluate reproducibility from different perspectives [64]. In fact, each dataset is evaluated as R, PR, or HR as follows:

$$R_t = \begin{cases} R & R_s \geq 0.75 \\ PR & 0.5 > R_s > 0.75 \\ HR & Otherwise \end{cases} \tag{2}$$

3.3. Datasets

N-BaioT (2018) [59]: A novel network-based dataset has been proposed for detecting botnet attacks in the IoT environment. The authors used two widely known botnets, Mirai and BASHLITE, to perform an attack on nine commercial IoT devices. Different statistical features were extracted over several temporal windows and then used by a deep learning autoencoder for attack detection.

IoTHIDS (2018) [47]: A host-based IoT dataset which captures data from real IoT devices. The experiment was conducted in a laboratory and consisted of three devices infected by Mirai, Hajime, Adira, BASHLITE, Doflo, Tsunami, and Wroba malware botnets.

IoT-SH (2019) [53]: This dataset was proposed using 8 different smart home devices on which 12 different attacks categorized using 4 main attack categories were performed. The authors followed a three-layer IDS to detect the attacks and used various combinations of rule-based and machine learning approaches to classify the attacks.

BoT-Iot (2019) [52]: This dataset is the representation of realistic traffic, which has been created containing heterogeneous network profiles. DDoS, DoS, data theft, and scan attacks are performed on five devices. A set of new features were extracted

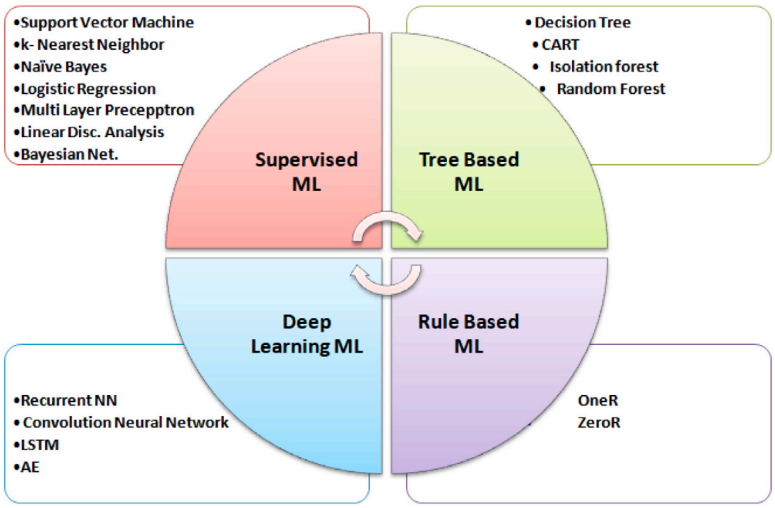

Fig. 6. Various machine learning and deep learning approaches used with IoT datasets.

using the Correlation Coefficient and Joint Entropy techniques. The features are then fed into various machine and deep learning methodologies for reporting the attack detection accuracy.

Kitsune (2019) [51]: An IoT dataset composed of four different categories of attacks performed on nine IoT devices. Among them, one IoT device (i.e., security camera) was infected by a real Mirai botnet sample. The authors presented Kitsune as a plug-and-play network intrusion detection system (NIDS) to detect normal and malicious samples.

IoTNIDS (2019) [60]: A dataset collected in a real environment recorded from two IoT devices (speaker and camera). The Mirai, MITM, DoS, and scanning attacks were performed on these devices.

MedBIoT (2020) [49]: A novel medium-sized IoT network architecture dataset is provided with labeled behavior. The dataset comprises seven IoT devices, some of which are real, and others are emulated. The authors used 100 statistical features passed into different machine learning techniques to detect the attacks.

IoT-23 (2020) [65]: A botnet dataset captured during 1 year that contains real environment captures of benign and malicious samples. This dataset was created as part of the Apsemat project that focuses on researching the attacks on IoT. During the period, authors used 23 different botnet attacks on the devices.

IoTIDs (2020) [23]: A dataset proposed with a set of IoT-related flow-based features, which were selected using the Correlation Coefficients technique and ranked with the Shapira–Wilk algorithm. For the study, the authors performed four different attacks on two different devices (speaker and camera) and recorded the data. They used SVM, G-NB, LDA, and LR for attack detection and classification.

MQTT (2020) [50]: The objective of the authors is to provide a dataset that uses a protocol dedicated to IoT network scenarios. Eight IoT devices were connected to the MQTT broker to represent it as real scenarios. A set of 33 different features were explored and fed into various machine learning algorithms to report the accuracy.

MQTT-IoT-IDS (2020) [62]: A dataset using a lightweight protocol, i.e., MQTT used in IoT network. The authors used a camera feed, 12 MQTT sensors, and a broker in the network. Their dataset consists of five scenarios, which include normal and attack data. The authors used packet-based, uni- and bi-flow features, inputted into six different machine learning algorithms to check the system's effectiveness.

TON-IoT (2020) [48]: A Telemetry-based data-driven IoT/IIoT dataset, which is heterogeneous in nature. The data captured consists of both normal and attack samples with ground truth. The dataset, which further represents realistic traffic, includes the sub-category of attacks. Apart from seven IoT devices, the dataset includes data recorded from operating system logs and network traffic. Various machine and deep learning algorithms are used, and they report the accuracy achieved.

Edge-IIoTSet (2022) [63]: a realistic cybersecurity dataset of IoT and IIoT applications to enable the development of intrusion detection systems in centralized and distributed applications. The authors present a in-depth description of the testbed used, as well as the dataset generation framework and considerations on centralized and federated learning. It is also focused on addressing the limitations of existing datasets.

4. Methodologies to detect IoT attacks

There is always a need to provide a robust environment to secure the IoT devices connected over a network. Various detection approaches like tree-based, classical supervised, rule-based and deep learning methodologies have been proposed over the years to detect these attacks. These algorithms provide promising solutions to the issues that arise [66]. Furthermore, different ML and DL techniques have been applied to IoT datasets. Fig. 6 depicts techniques categorized into tree-based, classical supervised, deep learning, and rule-based models. We have also provided information in Table 5 showing the number of features used, the type of features extracted, techniques used for feature selection and algorithms used to detect the attacks.

Table 5
Performance comparison of IoT datasets with features type, selection technique and methodologies.

Datasets	# of Features	Feature type	Feature Selection	Algorithms	k-fold	Accuracy	Precision	Recall	FAR	FPR	F1-Score
BoT-IoT	10	Flow	corr coeff, entropy	SVM,RNN,LSTM	5	0.88 (SVM)	1	0.88	–	0	–
Kitsune	115	Statistics	–	IF,GMM	–	–	–	–	–	–	–
N-BaIoT	115	Statistics	–	IF,LOF,SVM, AE	4	–	–	–	–	0.007 ± 0.01(AE)	–
Med-BIoT	100	Statistics	–	k-NN,DT,RF	10	0.97 (RF)	0.98	0.97	–	–	0.96
IoTID	83	Flow	corr coeff	SVM,LR,GNB, LDA,RF, DT	3,5,10	88 (DT)	88	88	–	–	88
IoT-SH	10	Packet	corr coeff	SVM,MLP, NB,RF, OneR,ZeroR, LR,J48,BN	10	–	0.99 (J48)	0.99	–	–	0.99
TON-IoT	44	Flow	–	LR,LDA, k-NN, CART, RF,NB, SVM,LSTM	4	0.77 (CART)	0.77	0.77	–	–	0.75
MQTT	33	Packet	–	NN,GB, NB,DT, MLP,RF	–	0.91 (RF)	–	–	–	–	0.91
MQTT-IoT-IDS	43	Packet & flow	–	LR,k-NN, SVM,RF, NB,DT	5	88.55 (DT)	88.55	88.55	–	–	88.54
Edge-IIoTset	63	Packet & flow	Feature Importance	DT, RF, SVM, KNN, NN	–	0.9467 (NN)	0.8019	0.7699	–	–	0.7733

4.1. Tree-based machine learning (ML) models

The most commonly used techniques are tree-based ML models, consisting of nodes and branches. Generally, a tree is built recursively, splitting the training samples by creating a node with the help of a feature that separates the data effectively. Some information criterion governs the selection of the feature. For predicting the label of the given test/query sample, it compares feature values from the root node and traverses down via different branches and nodes to reach the terminal node.

Decision Tree (DT) [67] is a technique widely used in detecting, classifying, and mitigating IoT attacks. Regarding Intrusion Detection Systems (IDS) for IoT, some contributions focus on a combination of DT with rule-based approaches [67] or hybrid models [68]. It has been used in simple and challenging problems since it is efficient and easy to train and interpret. Similarly, Random Forest (RF) [69] has presented outstanding result in several initiatives. Both in hybrid [70] or direct applications [71], RF has been adopted to classify and detect IoT threats in several systems. In fact, it refers to an ensemble of multiple DT where each tree is built by randomly selecting the data points.

Isolation Forest [72] is an unsupervised outlier detection algorithm based on the isolation of outliers and is useful in many cases, such as intrusion detection [73] and anomaly detection [74]. Finally, the random partitioning of features performed by this algorithm results in shorter tree paths for outliers in comparison to normal data points. Furthermore, Classification and Regression Trees (CART) [75] is a non-parametric, decision tree-based algorithm that constructs a binary tree structure from the training set. This technique has been used in several IoT-based security applications, e.g., IDS in smart grids [68] and anomaly detection SDN-based applications [76].

4.2. Classical supervised machine learning models

This is the most common branch in ML and is used for intrusion detection in IoT. Here, we describe some of the most commonly used methodologies in IoT applications.Among them, SVM, k-NN, and NB are mostly selected for reporting the results.

Support Vector Machine (SVM) [77] has been used in IoT privacy preserving [78] and in secure smart home applications [79]. This technique is based on a hyperplane in an N-dimensional feature space that optimally separates the classes and can be resource demanding in some cases. Similarly, Naive Bayes (NB) [80] has been successfully employed in several IoT security solutions since it does not require extensive datasets in order to produce good results. Examples of IoT applications based on NB are IoT-Fog-Cloud anomaly detection [81] and attacks detection in wireless sensor networks [82]. This scalable technique works with the a-priori assumption that the features are independent of each other.

IoT security applications can be supported by Linear Discriminant Analysis (LDA) [83] in different cases, such as in improving the performance of classifiers in intrusion detection [84] and IoT terminal recognition [85]. This method finds a new feature space to maximize the class separability while minimizing the within-class variability. In addition to classification, LDA is popularly used as a dimensionality reduction technique. Logistic Regression (LR) [71] is another popular approach used for binary classification problems where a non-linear sigmoid/logistic function governs the decision. It has been used in problems such as trust-based misbehavior detection [41] and intrusion detection [86], and maps an input to a value between 0 and 1 where a threshold value can be set to indicate the query sample be classified in one of the binary classes.

Multilayer Perceptron (MLP) [87] refers to a shallow variant of feed-forward neural networks with high performance in several problems. It has been used in solutions related to, e.g., trustworthiness [88] and attack detection [89]. k-NN [90] is a non-parametric

supervised machine learning approach commonly used for multi-class classification problems has presented remarkable performance in solutions of Intrusion Detection Systems (IDS) in Wireless Sensor Network (WSN) [91,92]. To be robust against noise and effective in working on large datasets, the output value is predicted by calculating the distance of the k-nearest neighbor from the core point.

4.3. Deep learning based approaches

Recently, with the required computing capability and huge datasets for model training, deep learning algorithms are heavily used across different applications due to supremacy in terms of accuracy. This section discusses variants of deep learning algorithms popularly used in literature.

Convolution Neural Networks (CNNs) [93] is an extension of the multilayer perceptron composed of an input layer, many hidden layers, and an output layer. The hidden layers are made by convolution, non-linear, down-sampled (i.e., pooling), and fully connected layers. This model has been successfully used in problems such as near real-time security [40] and malware analysis [94]. Furthermore, Recurrent Neural Networks (RNNs) [95] refer to a popular neural network architecture to model the sequential data due to their ability to remember the inputs with the help of their internal memory. In IoT security, RNNs have been used for several purposes, such as IoT IDS [96] and Networking Malware Threat Detection [97].

The Long Short-Term Memory (LSTM) [98,99] is an advanced variant of RNNs, which helps in learning long-term data dependencies. This was achieved with the help of memory known as gated cells, which have 3 gates: input, output, and forget gate. The functionality is flow control of information that needs to keep or delete to predict the output in the network. Finally, Auto Encoder [100] is a specific type of neural network which works in a feedforward direction. Here the output is the same as the input, where input is first compressed into low-dimensional form and then reconstructed at the output. This technique has been used in specific attacks, such as impersonation attack detection [101] and anomalous communication detection [102] in IoT.

4.4. Rule based machine learning approaches

In this type of machine learning, specific rules are defined to detect malicious activity. Whenever the data is processed, the system checks the defined attack behavior in the rule book and then reports what it detects.

OneR [103] is a depth-1 decision tree, which is simple and often used in intrusion detection. This algorithm generates a particular set of rules starting with a single attribute. This way, every branch is assigned the most frequent class. Similarly, ZeroR [104] is a simple rule-based classifier is ZeroR, which relies only on the target class and ignores all the predictor values. From the frequency table maintained, it never chooses the non-target value. Therefore, this classifier works best when baseline performance needs to be determined for other classifiers.

5. Potential features used in IoT datasets for attack detection

Over the years, various frameworks have been proposed for detecting network intrusions. But these proved shallow if a proper set of features is not provided. Although researchers have proposed different techniques to select appropriate features to recognize the type of attacks, we still lack the tools to determine what makes a good set of features. To detect attacks on individual host there are HIDS and to examine the abnormal behavior of the network different NIDS are available. The features are categorized into different groups that are described next:

5.1. Connectivity features

These features carry all the information related to source and destination. These features are further categorized according to their usability in extracting the information related to the attacks as *Basic* These are the features that usually contain the information regarding the source and destination flow like source–destination IP, protocol type, etc. These features are also defined as coarse-grained or network identifier features, are usually required for any attack detection. Examples of IoT applications based on these features are IoT device identification [105], intrusion detection [106], and malicious traffic detection [107]. *Time* refers to information regarding the duration of connectivity, activeness, idleness, or direction of the two flows is described in detail using time features. *Bytes* refers to checking the original number of the source to destination IP and total count of bytes sent in forward or backward direction. These features give essential information regarding attacks. *Flag* is about insights into how protocols like TCP and MQTT are subject to numerous attacks. The features provide information regarding the setting of flags in packets (either forward or backward) and the connection state.

5.2. Layered features

The increase of cyber attacks on IoT-related devices has focused on the type of communication protocols used at different layers. As protocols are the primary communication medium over the network, they are more vulnerable to attacks. To identify the features, protocols at each layer need to be identified and appropriately analyzed. Example of applications that use these features are fingerprint-based [108], anomaly-based [109], and traffic-based [110] solutions. The layers on which the intrusion happens include L4: Application Layer, L3: Transport Layer, L2: Network Layer, L1: Network Interface or Sensing Layer.

5.3. Communication features

IoT has become an extended concept, as it expresses a framework through which diverse objects, such as smart metering, smartphones, or wireless sensor nodes, are blended and connected via the internet through unique addressing. These devices interact and work in a coordinated way to achieve specific goals in a stipulated time frame. This diversity in technology has provided a fertile ground for threats to manifest [111]. The feature set includes features related *wifi* to where it should help define intrusion detection accuracy by minimizing processing and storage requirements [112], *bluetooth low-energy(BLE)* which is a well-designed connectivity protocol used by lower-powered devices and is an emerging choice in many IoT device applications and becoming an easy prey for cyber attackers and *zigbee* which is popular for use in IoT devices such as smart meters and home automation. This popularity resulted in the discovery of security vulnerabilities. The Zigbee frame provides crucial features that can help counter attacks on IoT devices and several works are using machine learning for this end [113–115].

5.4. Dynamic features

These features are essential for analyzing the behavior of the information flow. These include traffic statistics, such as the packet size, original IP addresses, and inter-arrival time. The dynamic features are useful, as they reveal information about deviation from the norm of traffic patterns to identify attacks [116]. In [117], authors used calculations of flow features, such as mean, max, and std to propose an intrusion detection system. Similarly, a novel software-defined network-based IDS framework is developed, and statistics features are used for attack categorization. The connectivity and layered features used by IoT datasets for attack detection are presented in Fig. 7(a), and flag features are depicted in Fig. 7(b) and dynamic features are shown in Fig. 7(c). In fact, these features are vital for ML models once they provide information about the underlying characteristics of the traffic and enables such techniques to identify patterns or normal and malicious network activities.

6. Key challenges & conclusion

IoT has become an essential commodity in our day-to-day lives. With its multifaced capability to cater to users' needs, it has become a personal assistant that everyone is interested in using. However, the popularity of IoT has attracted intruders who hack the system and take control by stealing user credentials or sending multiple requests to break down the system. Subsequently, the research community is working towards proposing realistic/simulated datasets and providing suitable solutions based on specific scenarios. This work has identified ten parameters for performing comparative analysis among IoT datasets. However, we need to discuss some critical aspects related to the publicly available datasets.

- **Device Selection and Functionality:** The datasets proposed include smart IoT devices, such as baby monitors, doorbells, and security cameras. This technology is changing quickly, and the embedded techniques, functionalities, criteria, and specifications are obsolete or will become obsolete with time. During this study, we also observed that some of the devices used in the proposed datasets have either single (camera) or multiple functionalities (baby-monitor). This raises a concern that an attacker has an advantage in multi-functional systems because of the many ways of entering the system without being noticed. Therefore, we hope that future datasets will include the latest up-to-date devices so that any new attack can be easily identified.
- **Traffic Format:** Among the datasets discussed, only a few have exposure to real-world data in a simulated environment. Also, the tools to collect such data vary, i.e., there is a lack of following a particular standard format, which affects the usefulness of these intrusion detection systems.
- **Attacks and Variants:** During this survey, we noticed that some of the proposed datasets used a main category of attacks while others used subcategories of attacks. This creates a dilemma about whether to provide a solution for the main class or a subcategory of an attack. As the proposed systems need to maintain a view of real-world scenarios, it is best to know both categories and circumstances.
- **Perfect Methodology:** New approaches are being proposed to detect attacks on the system. During our study, we found that no perfect approach could help identify a particular attack. Because one attack can be detected by more than one approach or by a combination of machine or deep learning approaches, one or more approaches must be used to detect one or more types of attacks. Therefore, we recommend a deeper investigation towards proposing better solutions.
- **Security updates:** IoT manufacturers are more eager to produce and deliver their devices as fast as they can, without giving security too much thought. Also, most manufacturers offer only firmware updates.

With the ever-increasing demand for IoT devices in various domains, such as health, industry, and education, IoT environments offer potential ground for intruder attacks. In contrast, the vulnerabilities of devices threaten user security and privacy. A robust solution for IoT must be developed to cover the properties of a wide range of attacks, including zero-day attacks. This paper presents an extensive survey of state-of-the-art approaches to intrusion detection IoT datasets by using protocol description, attacks, vulnerabilities, feature description, detection methodologies, and accuracy, and by showing whether the compared surveys review IoT datasets. We have discussed security attacks in IoT that threaten various layers in the network framework, highlighting in-depth attacks performed at the layers. However, these attacks are performed on different protocols working at multiple layers, so we also discussed the attacks on TCP/IP protocols suite with exploited vulnerabilities. Further, we have presented publicly available IoT datasets in detail, exploring the type of data format used, whether the data is labeled or not, the duration of the data recorded, type of network, which country the experiment was performed, the number of devices used in their experiment and the number of benign and malicious samples collected. We then discussed various deep and machine learning methodologies used to detect attacks on IoT devices. We also enlisted potential features which would help detect unexpected behavior in IoT devices.

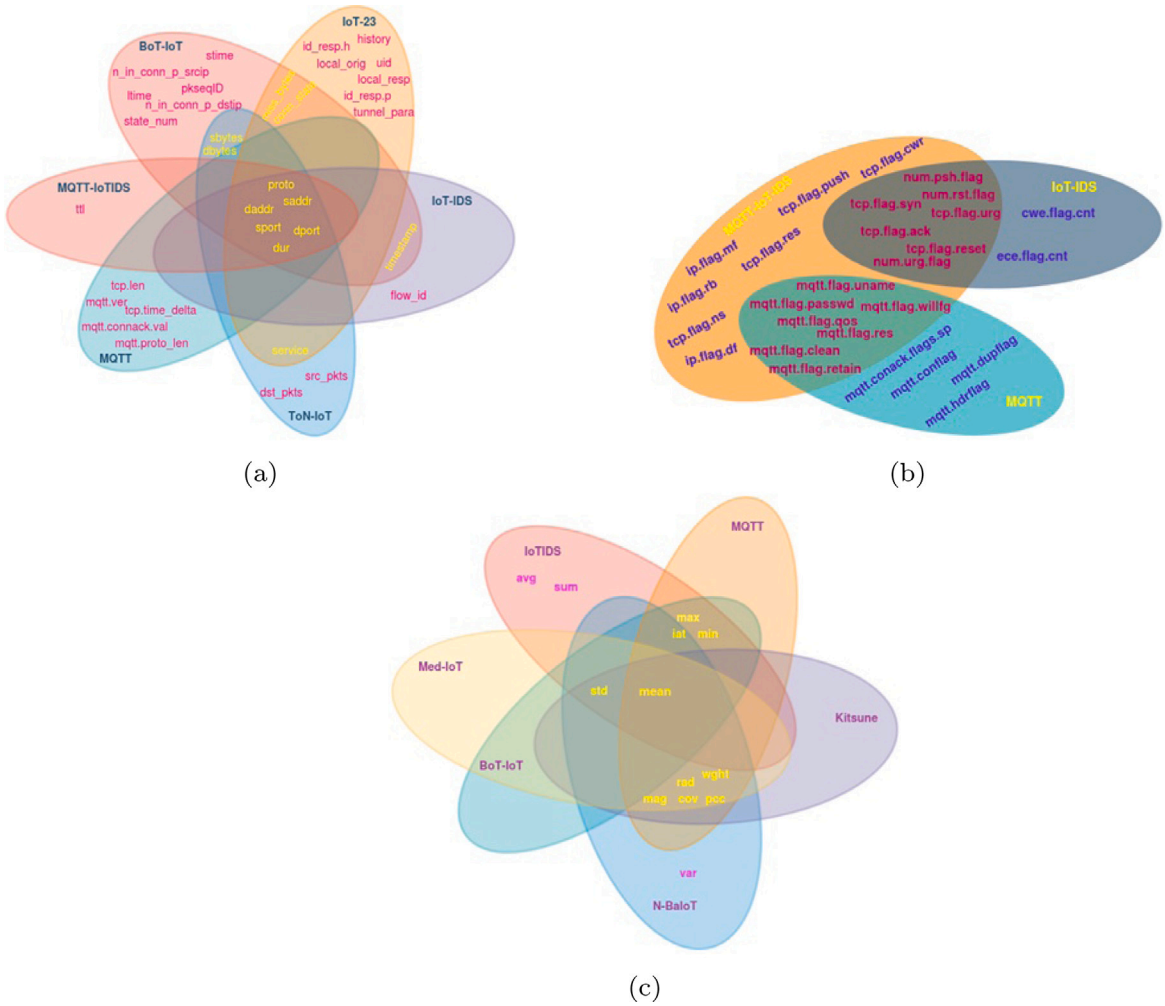

Fig. 7. A venn-diagram representing various features used by different IoT datasets.

7. Additional tables

In order to provide further insights on the characteristics of the datasets investigated in this research, we present a description of attacks on IoT at different layers (Table 6), a detailed summary of IoT dataset properties (Table 7), a list of potential features (Table 8), and a list of features used in IoT datasets (Table 9).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Table 6
Description of attacks on IoT at different layers.

Layer name	Attack type	Attack name	Description	Ref	Violation	Level of impact	Threat type
Physical/ Sensing	DoS	Tampering	These attacks generally focused on hardware components	[118]	A	H	S
		Jamming	Occupies the communication channel between the nodes thus preventing them from communicating with each other	[118]	A	H	S
		Malicious Code injection	Constantly sending the malicious messages to block or overload the network	[118]	C	M	P
		Battery Exhaustion	Attack by sending large number of request	[34]	A	H	S
	Physical Damage	Power loss	Uninterrupted power loss	[58]	A	H	S
	Personal Identification	Social Engineering	Using personal information from IoT gadgets to gain illegal access	[119]	C	H	P
	Data transit	Eavesdropping	Reading messages by unintended receiver	[120]	I	M	S
MAC/Network/Transport	DoS/DDoS	Selective Forwarding	To block the messages from forwarding and malicious user acting as a black hole this will prevent from messages to further propagate	[121]	I	H	P
		POD	Attack performed by making the packet size invalid ultimately destroying the target device.	[122]	I	M	S
		Flooding	Blocking a machine or network by sending invalid information.	[123]	A	H	S
		TCP Flood	Sending overwhelming number of ping requests	[59]	A	H	S
		UDP Flood	A numerous packets sent to random ports in this attack	[23]	A	H	S
		Smurf	Consume enough server resources to and making unresponsive system to legitimate traffic	[124]	A	H	S
		ICMP	A large number of ICMP message are sent to crash the system.	[125]	A	H	S
		Telent	Unavailability of Telnet services for remote administration	[51]	A	H	S
		Sybil	Replicating the single node and presenting it using multiple identities	[55]	I	H	P
	Spoofing	ARP spoofing	Changing the MAC and IP address	[126]	I	M	S
		MAC spoofing	Used by the attacker to present as legitimate IoT device	[127]	I	H	S
		MAC flooding	Interrupt normal flow of traffic between sender and receive	[20]	I	H	S
	MITM	Eavesdropping	Attacker collect information from a network by snooping on transmitted data	[128]	I	M	P
	Hole attacks	blackhole	Attack is to drop the incoming data packets	[129]	I	M	S
		sinkhole	Making the victim node attractive so that all flow goes towards victim node thus making it prone to attack	[55]	A	M	S
		wormhole	Entraps the packets and send it to far off node inside or outside the network	[130]	A	H	S
		greyhole	Depletion of the resources and causing the overhead to the network is the motive of the attacker	[131]	A	H	S
Application	Phishing	social engineering	Disclosure of username, password and confidential Information	[132]	C	H	P
	Injection	SQL Injection	Gaining unauthorized access to several websites	[132]	C	H	P
	Malware	virus,worms	Disrupt or intercept the legitimate confidential data.	[58]	C	H	P
	DoS	SMTP	Resource starvation	[133]	A	H	P
		Buffer overflow	An attack performed by writing a long sequence of malicious code	[134]	C	M	P
	DDoS	Amplification	The attacker forge IP addresses of the CoAP nodes to perform amplification	[135]	C	H	P
	MITM	Replay	Here the attacker store the information about the user without its consent or knowledge	[136]	C	M	P
Data Layer	SCADA	SQL Injection	Most common attack of injecting the SQL code into the system thus slowing the services	[137]	C	H	P
	Malware	Virus, Worms, Trojan,spyware	An adversary can damage the system by using malicious code which spreads to attached emails	[58]	C	H	P
		Eavesdropping	Reading messages by unintended receiver	[120]	C	M	P
	Cryptanalysis	Ciphertext	The ciphertext is easily accessible also the corresponding plaintext	[58]	C	L	P
		Plaintext	Some parts of the plaintext is known to the attacker and it modifies it	[58]	C	H	P

Violations (C), (I), (A) mean Confidentiality, Integrity and Availability.
Level of Impacts (H), (M), (L) mean High, Medium and Low.
Threat Types (S), (P) mean Security, Privacy.

Table 7
A detailed summary of IoT dataset properties.

Dataset	Device type	Simulation method	Commu.type	Data Capturing Tool	Attack type	Attack Name	Attack tool	Attack samples
N-BaIoT	doorbell-Danmini	n.s.	wifi	wireshark	Bashlite	Scan	binaries	255,111
(2018)	doorbell-Ennio					Junk		261,789
	Thermostat-Ecobee					TCP		859,850
	babymonitor-Philips B120N/10					UDP Flooding		946,366
	security camera-Provision PT-737E					COMBO		515,156
	security camera-Provision PT-838				Mirai	Scan	binaries	537,979
	security camera-SimpleHome XCS7-1002-WHT					Ack		643,821
	security camera-SimpleHome XCS7-1003-WHT					Syn		733,299
	webcam-Samsung SNH 1011 N					UDP Flooding		1,229,999
						UDP Plain		523,304
BoT-IoT	weather station	node-red	n.s.	tshark,	Information Gathering	service scanning	nmap, hping3	1463364
(2019)	smart fridge			argus		OS fingerprinting	xprobe3,nmap	358275
	Motion activated lights				Denial of service	DDoS	hping3,golden eye	38532480
	garage door					DoS	hping3,golden eye	33005194
	smart thermostat				Information Theft	Keylogging	Metasploitable3	1469
						Data Theft	Metasploitable3	118
Kitsune	Themostat	n.s.	wifi	NFQueue,	Reconnaissance	OS Scan	nmap	1697851
(2019)	baby monitor			afpacket,		Fuzzing	sfuzz	2244139
	web cam			tshark	Man in the Middle	Video Injection	video jack	2472401
	doorbell					ARP MitM	ettercap	2504267
	doorbell					Active Wiretap	raspberry pi 3B	4554925
	security camera - SNCEM602RC				Denial of Service	SSDP Flood	saddam	4077266
	security camera -SNCEM600					SYN DoS	hping3	2771276
	security camera-SNCEB600					SSL Renegotiation	THC	6084492
	security camera-SNCEB602R				Botnet malware	Mirai	Telnet	764137
IoTNIDS	speaker-SKT NGU (NU 100)	n.s.	wifi	n.s.	Mirai	UDP Flooding	binaries	949284
(2019)	Camera-EZVIZ WiFi (C2C Mini O Plus 1080P)					HTTP Flooding		10646
						ACK Flooding		75632
						Telnet Bruteforce		1924
						Host discovery		673
					MITM	ARP Spoofing	Nmap	101885
					DoS	SYN flooding	Nmap	64646
					Scanning	port	Nmap	20939
						os		1817
						host dicovery		2454
MedBIoT	Smart Switch-Sonoff Tasmota	raspberry PI	n.s.	Tcpdump	Botnet	Mirai	binaries	842674
(2020)	Smart Switch-TPLink					Bashlite		4143276
	Light Bulb-TPLink					Torii		319139
	Lock							
	Switch							
	Fan							
	light							
IoT-23	smart LED lamp-Philips HUE	n.s.	n.s.	wireshark	Botnet	Mirai	binaries	235,332,146
(2020)	Personal Assistant-Amazon echo home					Torii		100,000
	Doorlock-Smofy					Trojan		24000
						Gagfyt		271000000
						Kenjiro		109,000,000
						Okiru		13,000,000
						Hakai		23,000
						IRCBot		73,000,000
						Linix_Hajime		6,437,000
						Muhstik		496,000
						Hide & Seek		1,686,000
IoTID	speaker-SKT NGU (NU 100)	n.s.	wifi	CIC	DoS	SYN flooding	n.s	59391
(2020)	Camera-EZVIZ WiFi (C2C Mini O Plus 1080P)			Flowmeter	Mirai	Host Bruteforce		121181
						ACK Flooding		55124
						HTTP Flooding		55818
					MITM	UDP flooding		183554
						ARP Spoofing		35377
					Scan	Host Port		22192
						OS		53073
MQTT	Temperature	IoT-Flock	MQTT broker	wireshark	DoS	SlowIte	n.s.	9202
(2020)	Light intensity					Flooding	MQTT-malaria	613
	Humidity					MQTT Publish flood	IoT-Flock	130223
	Motion sensor					Malformed Data	MQTTSA	10924
	CO-Gas				MITM	Bruteforce	MQTTSA	14501
	Smoke							
	Fan speed controller							
	Door lock							
MQTT-	camera	VLC	n.s.	tcpdump.	Scan	Aggressive	nmap	70768
IoT-IDS						UDP		210819
(2020)					Bruteforce	Sparta	MQTT-PWN	947177

(continued on next page)

Table 7 (*continued*).

Dataset	Device type	Simulation method	Commu.type	Data Capturing Tool	Attack type	Attack Name	Attack tool	Attack samples
TON_IoT (2020)	Mobile-iPhone 7	Node-RED		wireshark, bro	Scan	Port	Nmap	25000
	Smart TV				DoS	DDoS	UFONet	3973
	Kali VM					Backdoor	Metasploitable3	35000
	Fridge				Malware	Ransomware	Metasploitable3	16030
	Garage door				Web	Data injection	bash scripts	35000
	GPS tracker					XSS	(XSSer) toolkit	6116
	Modbus				Information theft	Password cracking	CeWL,Hydra toolkit	35000
	Motion Light					MITM	Ettercap toolkit	1403
	Thermostat							
	weather							
Edge-IIoTset (2022)	Smart TV	N.A.		Wireshark	DoS/DDoS	TCP SYN Flood	hping-3	2020120
	Wireless Router					UDP Flood	hping-3	3201626
	Rapberry Pi 4 Model B					HTTP Flood	GET request	229022
	ESP32					ICMP Flood	hping-3	2914354
	Arduino Uno				Information Gathering	Port Scanning	Nmap and Netcat	22564
	Sensors & Actuators					OS Fingerprinting	xprobe2	1001
	Desktop Computers					Vulnerability Scan Attack	Nikto	145869
	Smart Phone				Man in the Middle (MITM)	DNS/ARP Spoofing	Ettercap	1403
					Injection Attacks	XSS	XSSer	15915
						SQL	sqlmap	51203
						Uploading	Metasploit	37634
					Malkware	Backdoor	Metasploit	24862
						Password Cracking	CeWL	1053385
						Ransomware	OpenSSL	10925

Table 8
List of potential features.

Category		Name	Description	Profile	Effectiveness
CF	Basic	Source IP	Indicates the address of origin IP	H	N
		Destination IP	Describes the destination IP address	H	N
		Source Port	The number of source port is stored in this feature	H	H
		Destination Port	It stores the port number of destination and available in TCP stack	H	H
		Protocol Type	Shows the type of protocol e.g. TCP,UDP etc.	H	N,H
	Time	Duration	Record total duration or flow duration	H	N,H
		Jitter	Source and destination jitter time in msec	L	N,H
		Inter Arrival Time	(avg,std,max,min) inter arrival time between two flows either in backward or forward direction	L	N,H
		Active Time	Time during which flow was idle before becoming active	L	N,H
		Idle Time	Time during which flow was active before becoming idle	L	N,H
	Flag	ACK, SYN,FIN,URG,RST	(fwd/bwd)Flag setting in the packets traveling in either direction	H	N,H
	Bytes	Count	Indicate the information regarding total number of unique source or destination IP bytes	M	N
		Header Length	Describes the information regarding total bytes used for headers in forward or backward direction	H	N
LF	L4	MQTT	This is a lightweight protocol and supports various authentication mechanisms built upon TCP	L	N,H
		CoAP	It is an open web transfer protocol designed especially for constrained devices and built on UDP	L	N,H
		HTTP	Used to deliver data on www	H	N,H
		HTTPs	Extension of HTTP , used for secure communication	H	N,H
		DNS	Translation of domain names into IP addresses	M	N,H
		Telnet	To provide bidirectional interactive communication	L	N,H
		SMTP	Suitable for mail transfer	L	N,H
		SSH	Used for remote logins	L	N,H
		IRC	Provides communication in the text format	M	N,H
	L3	TCP	It indicates connection oriented protocol and describes about 3-way handshake for full communication	H	N
		UDP	It is a connection less protocol and used for establishing low latency connections among devices	H	N
	L2	DHCP	It is used to dynamically assigns the IP addresses to the devices on the network	L	N
		ARP	Used for discovering link layer addresses such as MAC address	M	N
		RARP	It is the reverse of ARP	L	N
		ICMP	Used by networking devices to send error messages	H	N
		IGMP	Allows the network to direct multi-cast	M	N
		IPv(4,6)	It is core communication protocol with different versions 4 and 6	H	N
	L1	MAC	Provides a channel of access and address mechanism	M	H
		LLC	Can be used for packets on Ethernet networks	M	H
Cm F	wifi	Type	field indicates 0-management, 1-control, or 2-data	H	N
		Subtype	indicates the type of management, control, or data frame	H	N
		DS status	Indicates directionality of the frame	H	N
		Fragments	Set to 1, the frame has been fragmented and has more fragments to transmit	H	N
		Duration	reserve the medium for the amount of time	H	N
		Protocol Version	Describes which version is used.	H	N
		Src/dst	MAC address of the node that initiated the frame and final destination	H	N
		receiver addr.	MAC address that identifies the wireless device that is the immediate recipient of the frame	H	N
		Sequence number	Indicates the sequence number assigned to the frame	M	N

(*continued on next page*)

Table 8 (*continued*).

Category		Name	Description	Profile	Effectiveness
	BLE	Type	integer value corresponding to the packet	H	N
		Direction	selecting the direction of the packet if sent by gateway it is zero and received is 1.	H	N
		Timestamp	The packet's timestamp in epoch format	H	N
		Length	The packet's length in bytes.	H	N
		Taxonomy	Indicates whether the packet is management or data related	M	N
		Parameter Length	Extracted in case of HCI command type packets describing length of all the parameters measured in bytes	L	N
		Data length	The length of the packet's data in bytes in case of ACL data packets	L	N
		Event code	The code related to the different types of events described in HCI event packets	L	N
	Zigbee	Source address	The source address.	H	N
		Destination address	target information.	H	N
		Destination PAN ID	PAN Id information.	H	N
		Extended PAN ID	It is a default coordinator MAC address	M	N
		Timestamp	packet timestamp	H	N
		Packet length	packets byte length	H	N
		Data length	payload information in bytes	H	N
		Data	payload data in raw form	H	N
DF		Mean	Indicate average of a number	H	N,H
		Min	Minimum length of a packet	H	N,H
		Max	Maximum packet length	H	N,H
		Std	calculating std length	H	N,H
		IAT	Inter arrival time between packets	M	N,H
		Tot size	Size of the packet in bytes or bits	M	N,H
		Tot sum	packets byte count	L	N,H
		Avg	Average size of a packet	L	N,H
		Magnitude	The root squared sum of the two streams' means	M	N
		Radius	The root squared sum of the two streams' variances	L	N,H
		Co-variance	An approximated co-variance between two streams	H	N,H
		Variance	Ratio between outgoing and incoming traffic	H	N,H
		Number	Defines packet count	L	N,H
		Pearson Correlation	An approximated correlation coefficient between two streams	H	N
		Weight	Number of items observed in recent history	L	N,H

Connectivity Features (CF), Layered Features (LF), Communication Features (Cm F), Dynamic Features (DF)
Profiles (H), (M), (L) mean High, Medium and Low respectively.
Effectiveness (N), (H) mean Network Intrusion Detection , Host Intrusion Detection system respectively.

Table 9
List of Features used in IoT datasets.

Category		Name	BoT-IoT	Kitsune	N-BaIoT	MedBIoT	IoTID	IoT-23	IoT-SH	TON-IoT	MQTT	MQTT-IoTIDS	Edge-IIoTset
CF	Basic	Source IP	✓	✓	✓	✓	✓	✓		✓		✓	✓
		Destination IP	✓				✓	✓		✓		✓	✓
		Source Port	✓				✓	✓		✓		✓	✓
		Destination Port	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓
		Protocol Type	✓	✓	✓	✓	✓	✓		✓		✓	✓
		State	✓					✓		✓	✓		
		Service						✓		✓			
	Time	Duration	✓				✓	✓		✓	✓	✓	
		Jitter	✓	✓	✓	✓							
		Inter Arrival Time					✓						✓
		TTL							✓			✓	
	Flag	ACK, SYN,FIN,URG,RST					✓				✓	✓	✓
		count					✓						
	Bytes	Count	✓				✓	✓		✓			
		Header Length					✓						✓
LF	L4	MQTT									✓	✓	✓
		CoAP									✓		
		HTTP							✓	✓			✓
		HTTPs											
		DNS							✓	✓			✓
		BOOTP							✓				
		NTP							✓				
	L3	TCP							✓				✓
		UDP							✓				✓
		SSL							✓	✓			
	L2	DHCP							✓				
		ICMP							✓				✓
		IGMP							✓				
		IPv(4,6)							✓				
	L1	MAC	✓	✓	✓	✓							
		LLC											

(*continued on next page*)

Table 9 (*continued*).

Category	Name	BoT-IoT	Kitsune	N-BaIoT	MedBIoT	IoTID	IoT-23	IoT-SH	TON-IoT	MQTT	MQTT-IoTIDS	Edge-IIoTset
DF	Mean	✓	✓	✓	✓	✓					✓	
	Min	✓				✓					✓	
	Max	✓				✓					✓	
	Std	✓	✓			✓					✓	
	IAT					✓					✓	
	Tot size					✓						
	Tot sum											
	Tot num of bytes					✓						
	Tot num of packets					✓						
	Avg	✓				✓						
	Magnitude		✓	✓	✓							
	Radius		✓	✓	✓							
	Covariance		✓									
	Variance			✓	✓							
	Number											
	Pearson Correlation		✓	✓	✓							
	Rate	✓										
	Weight		✓									
Labeling	Category	✓	✓	✓	✓	✓		✓	✓	✓		✓
	Sub-category	✓	✓	✓		✓		✓		✓		✓
	Label	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓

Connectivity Features (CF), Layered Features (LF), Dynamic Features (DF).

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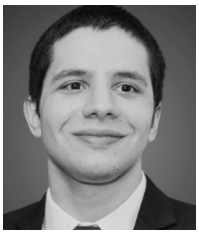

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